



Unit 4 · Outcome 3 · Scientific Investigation

Methodology & variables

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 02 Methodology & variables VCAA study design · the examinable dot point

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE CORE DISTINCTION

Methodology, method and technique are not the same word

Examiners lose patience with students who use these three as synonyms, because they describe three different levels of decision. Nail the hierarchy and a whole family of questions becomes easy.

The overall type of scientific investigation you choose — e.g. a controlled experiment or fieldwork. VCAA lists seven for Environmental Science.

The specific procedure that carries out that methodology — the ordered steps: what you do, to what, in what order, how many times.

The particular skill or tool used inside the method — e.g. using a quadrat, an EC (salinity) meter, or a dichotomous key.

◆ ANALOGY

Analogy · cooking dinner Methodology is deciding whether you bake, fry or grill — the whole approach. Method is the recipe: the ordered steps you follow. Technique is the knife skill — how you dice the onion. Choose to grill but follow a baking recipe and dinner is ruined, no matter how good your knife work is. In science, a mismatched methodology and question ruins the data the same way.

Priya, three levels: Methodology → controlled experiment. Method → grow 25 identical rush seedlings, water five groups with solutions of increasing salt concentration for 21 days, measure height every 3 days. Technique → make up solutions by mass with an electronic balance; measure height with a steel ruler to the nearest millimetre.

● NOTE

Why "characteristics of the methodology" earns marks When a question says "justify the chosen methodology" or "describe a characteristic of a controlled experiment," it wants the defining feature — e.g. a controlled experiment deliberately manipulates one variable while holding others constant so a cause-and-effect relationship can be tested. Naming the methodology is not enough; you must state what makes it that methodology.

LIVE MODEL · METHODOLOGY SWITCHER

One creek, seven ways to investigate it

Priya's worry — "salinity is harming the plants at Mirrin Creek" — can be turned into any of the seven VCAA methodologies. Pick one below: the model rebuilds to show how that investigation would actually run, and the panel gives its defining characteristic, when to choose it, and how variables behave in it.

● NOTE

The five you'll meet most in Environmental Science All seven are examinable, but in practice EnviroSci investigations lean on controlled experiments, fieldwork, correlational studies, classification & identification, and simulations/modelling. The deciding rule from VCAA: there must be congruence between the question and the methodology — the question you ask should determine the methodology you choose, not the other way round.

LIVE MODEL · THE VARIABLE ISOLATOR

The holy trinity: independent, dependent, controlled

A variable is any factor that can change or be changed in an investigation. In a controlled experiment there are three jobs a variable can have. Get the three straight and most of this dot point's marks are yours.

The single factor you deliberately vary across groups. The presumed cause. There should be exactly one.

The factor you measure to see if it responds. The presumed effect. It depends on the IV.

Every other factor that could affect the DV, held the same across all groups so only the IV differs. Usually several.

Below, drive Priya's experiment yourself. Slide the salt up (the IV) and watch the rushes' height respond (the DV). Then switch off a controlled variable and watch the result turn into noise — that's why CVs exist.

Slide the salt up: with all CVs held, growth falls cleanly as salinity rises — a clear dose–response. Turn a CV off and the bars scatter: you can no longer tell whether salt or the loose variable caused the change. That lost certainty is exactly what validity means.

WHERE MARKS ARE WON AND LOST

Operationalise: a variable isn't done until it has units

To operationalise a variable is to state exactly how it is changed or measured — with units, a range and (for the IV) the levels tested. Vague variables can't be marked because nobody could repeat the investigation from them. This single habit lifts answers from a 1 to full marks.

• NOTE

A quick test for any DV Ask: "What number, in what unit, will I write in my results table?" If you can't answer, your DV isn't operationalised yet. "Health" → mean shoot height (mm) or chlorophyll content or % leaves green. "Pollution" → electrical conductivity ($\mu\text{S}/\text{cm}$). Make it measurable.

THE MOST-CONFUSED PAIR IN THE COURSE

A control group is not a controlled variable

They sound identical and mean opposite things. Mixing them up is one of the most common ways students drop marks on this dot point, so here it is side by side.

◆ ANALOGY

Analogy · the mixing desk Picture a sound desk. The controlled variables are all the faders you tape down so they can't move — bass, treble, reverb. The control group is the "before" recording you keep so you can hear what your one moving fader actually changed. Tape down the wrong faders and you can't trust the mix; lose the "before" track and you can't prove anything changed at all.

Memory hook: controlled variables are plural conditions you hold steady; the control group is a single baseline you compare against. Different jobs, different words.

THE NUANCE EXAMINERS REWARD

Only a controlled experiment lets you control the variables

The IV–DV–CV language was built for controlled experiments, where you manipulate the IV. In the field you usually can't. Knowing how variables behave in other methodologies is a higher-order mark — and a classic exam trap.

• NOTE

The trap, stated plainly If Priya walks the creek and records salinity and plant cover at ten existing sites, she is doing a correlational study / fieldwork, not a controlled experiment — she never set the salinity. So she cannot claim salt caused the dieback, only that higher salinity is associated with lower cover. Confusing correlation with causation here is a guaranteed lost mark.

ACTIVE RECALL · DRAG & DROP

Sort each statement into its job

◇ ACTIVE RECALL

From Priya's investigation, drag each card into the correct bin. On a touch screen, tap a card then tap its bin. Hit Check to mark — green is right, red is wrong.

READ THE QUESTION PROPERLY

Command-word decoder

◆ COMMAND WORDS

The verb in the question sets how much you must write. Click a command word to see what it demands on a methodology-and-variables question, and a worked Mirrin Creek example.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

LOSE-THE-MARK MOMENTS

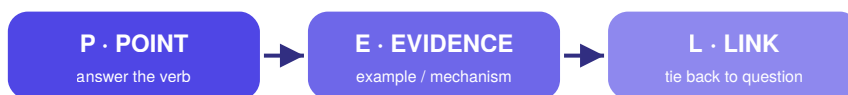
Seven traps that quietly cost marks

△ EXAM TRAPS

Each of these is a real pattern from examiner reports on investigation-design questions. Tap to open the fix.

BUILD THE PARAGRAPH

P·E·L trainer — justify a methodology



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Short-answer marks come from structure. P·E·L = Point (name and define) → Example (anchor in Mirrin Creek) → Link (tie back to the question). Type a justification below; the keyword meter lights up as you hit the ideas examiners look for.

EXAM-STYLE · 20 MARKS

Practice questions

★ PRACTICE & SELF-MARK

Five VCAA-style questions. For each, try it in your logbook first, then reveal the weak answer (and why it loses marks), the strong answer, and the marking guide. Mark yourself honestly — your score builds in the dock on the right.

• NOTE

Self-marking honestly is the skill. The dock totals the marks you award yourself across all five questions (out of 20). Aim to justify every mark against the guide, the way an assessor would — generous marking only fools you in November.

ONE-SCREEN RECAP

Take it into the exam

• NOTE

Built for the VCAA VCE Environmental Science Study Design 2022–2026, Unit 3 & 4 Outcome 3. Next in the chapter: KK03 — Generating primary data (qualitative vs quantitative techniques) and KK04 — Accuracy, precision & validity, which build straight on the variables you've just mastered. · Digital Vector · Australian English.

UNIT 3 & 4 · OUTCOME 3 · KEY KNOWLEDGE 02

Methodology & variables

"Characteristics of the selected methodology and method, and the appropriate use of independent, dependent and controlled variables."

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Not operationalised IV: salt DV: plant growth CV: conditions How much salt? Growth of what, measured how, over how long? "Conditions" names nothing. Unrepeatable → unmarkable.

✓ STRONG / MODEL ANSWER

Operationalised IV: NaCl concentration of solution: 0, 4, 8, 12, 16 g/L DV: mean increase in shoot height (mm) over 21 days CV: light (12 h/day), 22 °C, 50 mL water/pot, same potting mix, same species & age Levels, units and a time-frame mean another scientist could reproduce it exactly. That's the bar.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.